

Practical of Advanced Web Technologies

(RPSCSOP602)

A **12-hour, hands-on** course for Master's students covering the modern JavaScript full-stack: **Node.js, Express, MongoDB, TypeScript, and React.**

Every module is practical-first: you write and run code. Concepts are introduced only when needed to understand the next piece of code.

Course Outcomes (CO)

CO	You will be able to...
CO1	Demonstrate the working of Node.js
CO2	Apply the MVC pattern in programs
CO3	Classify code based on React, Node & modify based on database
CO4	Design database schemas and perform CRUD operations using MongoDB
CO5	Combine frontend and backend to build complete apps with seamless data flow

12-Hour Schedule

#	Module	Time	Practical(s)	CO
0	Setup & Tooling	0.5 hr	—	—
1	Node.js Fundamentals + Event Loop	1.5 hr	P4	CO1
2	HTTP & Express (status/headers)	1.5 hr	P3	CO1
3	MongoDB Data Models	1.0 hr	P1	CO4
4	CRUD with MVC	1.5 hr	P2	CO2, CO4
5	TypeScript	1.5 hr	P8, P9	CO3
6	React: First App & Components	1.0 hr	P5	CO3
7	React: Data Lists & CSS	1.0 hr	P6	CO3
8	React: Forms, Props & State	1.0 hr	P7	CO3
9	Full-Stack App (combine all)	1.5 hr	P10	CO5
	Total	12 hr		

Syllabus Practical → Module map

Practical	Description	Module
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P1	Implement MongoDB data models	3
P2	Simple CRUD app with Node.js + MongoDB	4
P3	Send various HTTP status codes and response headers	2
P4	Event loop and its role in async tasks	1
P5	React first app + Hello World with components	6
P6	Components to display a set of data + CSS	7
P7	Form in React using component props & state	8
P8	Data structures and functions using TypeScript	5
P9	Classes and inheritance (TypeScript)	5
P10	Full-stack web application combining all	9

How to use this repo

Each module folder contains:

- **README.md** — the concept + worked examples (this is what becomes the PDF).
- **code/** or **src/** — runnable, heavily-commented source code.
- **practice.md** — practice questions (and mini-tasks) for students.

Recommended path

1. Read the module **README.md** .
2. Open the code, run it, break it, fix it.
3. Do the **practice.md** questions before moving on.

Prerequisites

- Basic JavaScript (variables, functions, arrays, objects).
- A computer with **Node.js 18+** and a code editor (VS Code recommended).
- MongoDB — either installed locally or a free **MongoDB Atlas** account.

See [Module 0](#) for full setup instructions.

Generating the PDFs

All module notes can be exported to PDF:

```
npm install      # once, installs the md-to-pdf tool
npm run pdf      # generates pdf/*.pdf from every module README
```

PDFs are written to the **pdf/** folder.

License

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